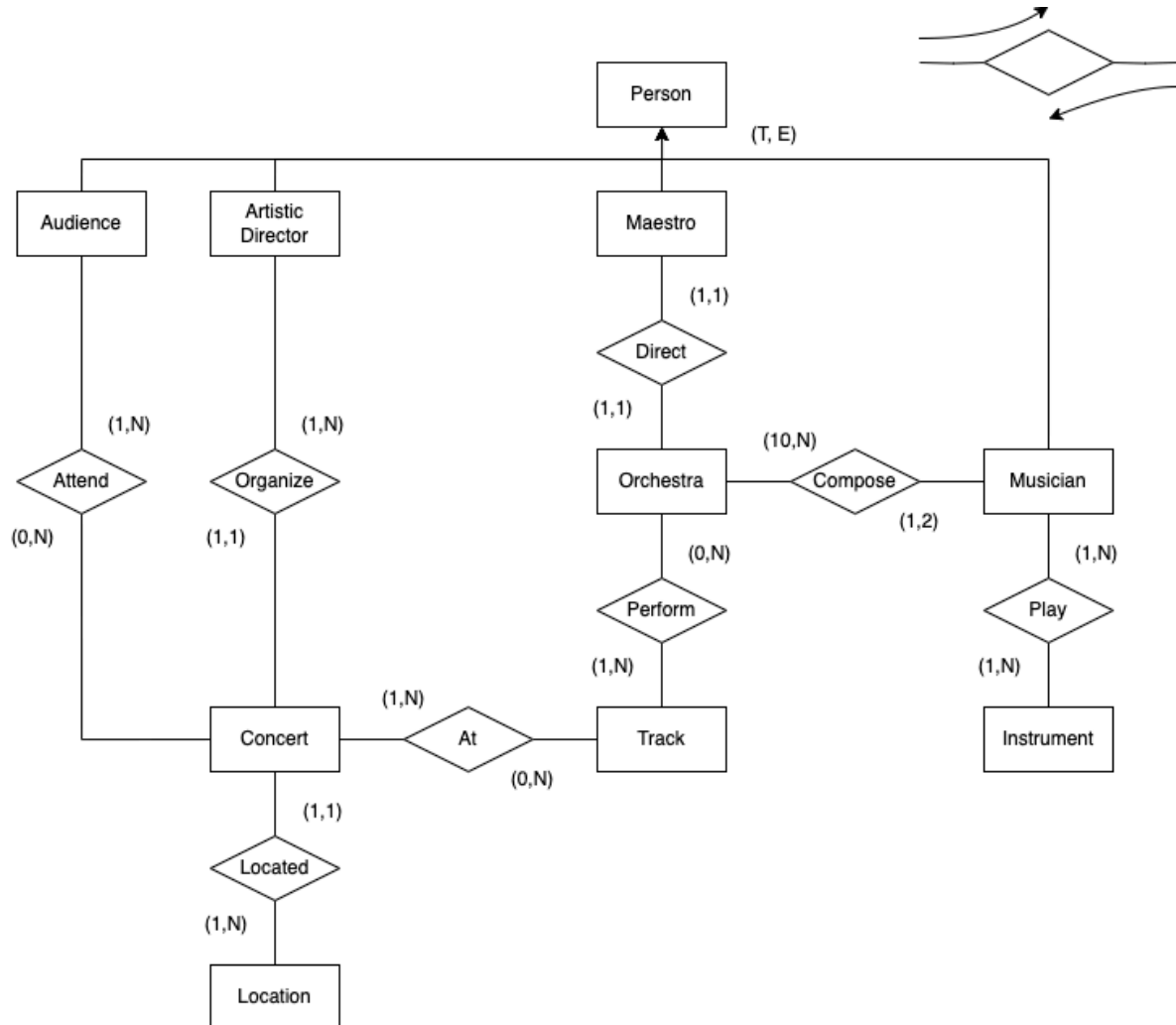


Last Name (Surname, Cognome) _____ First Name _____ Codice Persona _____

Consider the following ER Diagram, which represents a data structure for tracking concert attendances and performances.



The following attributes describe the entities. The primary keys are underlined.

- **Person** - Personal ID, Name, Surname, Birth Date
 - **Audience** - (...), Favourite Music Type
 - **Artistic Director** - (...), Biography
 - **Maestro** - (...), Biography, Stage Name
 - **Musician** - (...), Biography
- **Instrument** - Name, Type (e.g., *chordophones*), Description
- **Orchestra** - Name, Foundation Date, Biography
- **Track** - Title, Duration (minutes), Composer (e.g., *Mozart*), Music Type (e.g., *classical*)
- **Concert** - ID, Title, Start Time, Duration (minutes)
- **Location** - Name, Address, Type (e.g., *theater*), N Seats

The following attributes describe the relationships.

- **Play** - Since

- **Direct** - First Concert (Yes/No)

Exercise 1 - Unstructured data models (1 PT)

Which DB technology would be the best to store the entities/relationships of the ER diagram? [You may not need all the rows of the table] (1 PT)

ENTITIES / REL.	DB TYPE	MOTIVATION

Exercise 2 - Neo4j (5 PT)

Consider the entities Audience, Concert, Artistic Director, and Location from the ER model. Suppose you want to store the respective data instances in a graph database. Sketch a graph model/example describing the nodes, main attributes, and edges. Either show an example graph or a graph with types. (1 PT)

Write a Cypher query to count the total number of concerts managed by each artistic director whose name is "Chiara", that lasted more than 100 minutes, and with more than 1000 attendances(2 PT)

Write a Cypher query to collect, for each audience member that attended at least 30 concerts, the amount of different locations where the member participated in at least one concert. Consider only the people who attended at least 500 of such concerts and return their average age (*now()* - *birthdate* returns the age) (2 PT)

Exercise 3 - MongoDB (7 PT)

Consider the entities Orchestra, Maestro, Musician, and Instrument. How many collections would you define within a documental database (clearly state it)?

How would you implement the relations between the concepts? Provide a simple documental representation. (1 PT)

Write a query to compute the total number of musicians who can play the piano and whose name is "Fabiana" (1.5 PT).

Write a query to compute the number of members for each orchestra and each instrument, and return only the orchestras that include at least three musicians for at least four instruments (2.5 PT)

Write a query to count the number of orchestras directed by directors named "Claudio" and including at least one member named "Lorenzo" who can play the piano (2 PT)

Exercise 4 - Redis (3 PT)

Consider the entity Instrument.

Create a set of instruments including "Keyboard", "Violin", and "Drums" (1 PT)

Perform an operation to check whether “Flute” is part of the set (1 PT)

Perform an operation to add “Flute” to the set, with time to live equal to 10 (1 PT)

Exercise 5 - Elasticsearch (3 PT)

Consider the Track entity. Provide the complete mapping of the index for the considered entity (i.e., field name, field type, the structure of the mapping, etc.) (1 PT)

PUT /

Write a query to return the documents whose composer is “Bach”, whose duration is greater than 8 minutes, and whose music type is classical, prioritizing those with duration greater than 11 minutes. The condition on the composer must not affect the final score (2 PT)